

Income Inequality and Poverty

ECON 204 – Contemporary Economic Issues

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Winter 2026

- How to define and measure **income inequality** and **poverty**.

Learning Objectives

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- How to compare Canada with other countries.
- How to analyze the comovement between inequality, poverty, and **real per capita GDP**.

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- **Economic mobility:** This characterizes the ability for individuals to move from one part of the income distribution to another.
- **Gini coefficient:** This is a global measure of inequality. It goes from 0 (perfect equality) to 1 (perfect inequality).

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- **Low income measure (LIM):** This is equal to half the median of income. It is one of the measures of the poverty line. Individuals with income less than the LIM are considered below the poverty line.

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- **Median:** This is a number that separates the distribution of a variable in two equal groups.
- **Per capita:** This is a Latin word that means “by head”.
- **Quintile:** When the population is divided into five equal groups ordered with respect to a variable, the groups are called quintiles.

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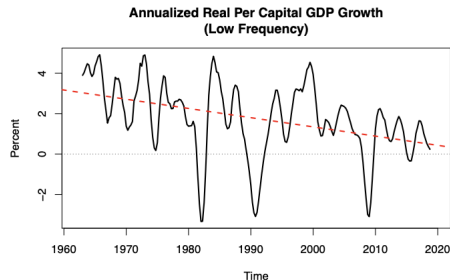
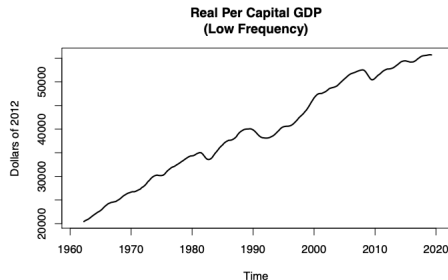
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 - the population series.
- Be careful with units: population is in *units* and GDP is in *millions*; express population in millions before dividing.

Real per capita GDP and its growth (illustrations)



- Low-frequency component of real per capita GDP (left) and annualized low-frequency real per capita GDP growth (right).
- Real per capita GDP rises from about \$20,000 to about \$55,000 (2012 dollars) between 1961 and 2020; growth rate declines on average.

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- But it does **not** imply that the standard of living of *every* Canadian has improved.
- This lecture focuses on the **allocation** of real GDP/income across individuals, how it evolves, and how it relates to growth.

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- The lowest quintile contains the 20% with the smallest values; the fifth quintile contains the 20% with the highest values.
- Which individuals fall into each quintile depends on the variable used (e.g., age vs expenditure).

Expenditure and age by individual

Individual	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Age	19	23	45	18	64	55	34	23	22	44	70	61	49	21	43
Expenditure	100	210	80	30	312	190	201	236	341	34	40	101	170	190	343

Age quintiles (sorted by age)

	First	Second	Third	Fourth	Fifth
Individuals	4, 1, 14	9, 2, 8	7, 15, 10	3, 13, 6	12, 5, 11
Age	18, 19, 21	22, 23, 23	34, 43, 44	45, 49, 55	61, 64, 70
Expenditure	30, 100, 190	341, 210, 236	201, 343, 34	80, 170, 190	101, 312, 40

Measuring inequality: income by quintile

- A simple way to summarize inequality is to compare the **average income** across quintiles and the **share of total income** held by each quintile.

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- A simple way to summarize inequality is to compare the **average income** across quintiles and the **share of total income** held by each quintile.
- If higher-income quintiles capture a much larger income share, the distribution is more unequal.

Average income by quintile in 2018

Quintile	Average Household Income (dollars)	Share of Total Household Income (%)
First Income Quintile	23,537	5.90
Second Income Quintile	49,137	12.40
Third Income Quintile	69,943	17.60
Fourth Income Quintile	91,798	23.10
Fifth Income Quintile	162,231	40.90

Cumulative income share in Canada in 2018

- The cumulative share of income is the share of income of the first quintile (bottom 20%), the first two quintiles (bottom 40%), the first three (bottom 60%), and so on.

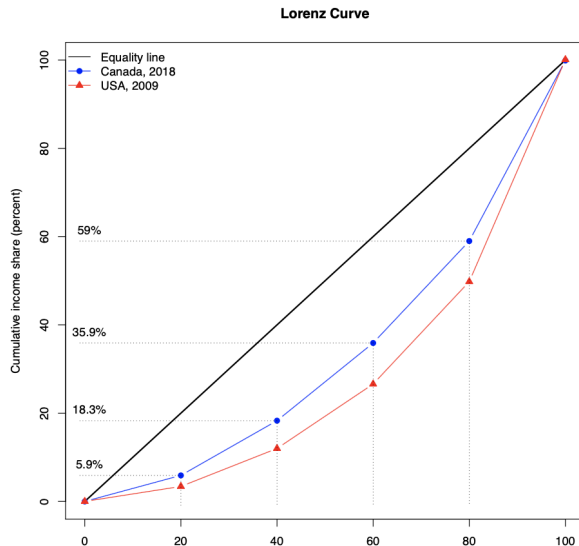
Group	Cumulative Income Share (%)
The bottom 20%	5.90
The bottom 40%	18.30
The bottom 60%	35.90
The bottom 80%	59.00
The bottom 100%	99.90

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- Perfect equality implies: bottom 20% gets 20%, bottom 40% gets 40%, etc. (the 45-degree equality line).
- The farther the Lorenz curve is from the equality line, the more unequal the distribution.

Lorenz curve: Canada (2018) and USA (2009)



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- Lorenz curve with equality line.
- Coordinates shown for Canada: 5.9%, 18.3%, 35.9%, and 59%.
- Conclusion: USA (2009) allocation is more unequal than Canada (2018).

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- It equals 0 when the Lorenz curve is the equality line, and increases as inequality rises.

Gini coefficient with quintiles

- Let $h_i = (i/5 - s_i)$ for $i = 1, 2, 3, 4$, be the distance between the equality line and the Lorenz curve at 20%, 40%, 60% and 80%, where s_i is the cumulative income share of quintiles 1 to i .

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- Then the Gini coefficient is:

$$\text{Gini} = 0.40 \times (h_1 + h_2 + h_3 + h_4) = 0.8 - 0.4 \times (s_1 + s_2 + s_3 + s_4).$$

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- With quintiles, the maximum Gini is 0.8 (to reach 1, we need more than 5 groups).

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- Add trapezoid areas across segments and multiply by 2 to get the Gini coefficient.

Gini coefficient in general (n groups)

- If we split the population into n equal groups, define $h_i = (i/n - s_i)$ for $i = 1, \dots, n - 1$, where s_i is the cumulative income share of groups 1 to i .

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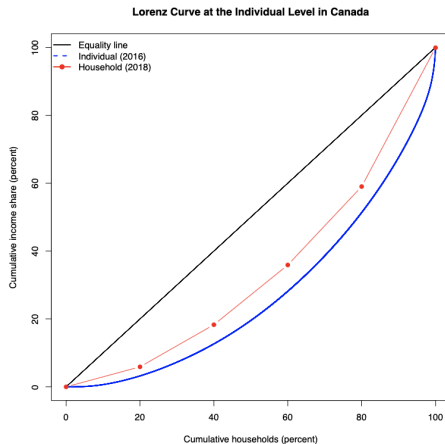
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- The maximum is $(n-1)/n$, which approaches 1 as n becomes large.

Lorenz curve: individual vs household



- Lorenz curve at individual level (2016) and household level (2018), with equality line.
- More groups $\Rightarrow$ smoother curve; inequality is higher at the individual level.

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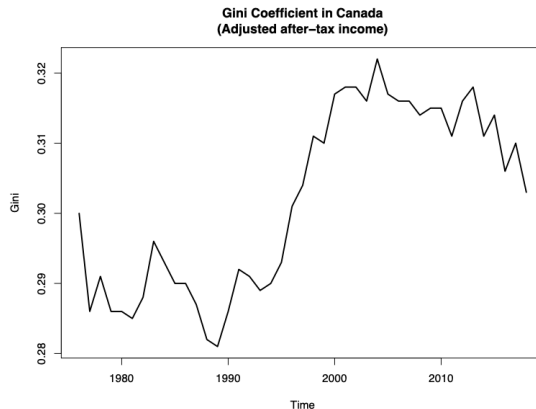
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- The business cycle affects inequality (e.g., increases during recessions of 1982 and 1990).

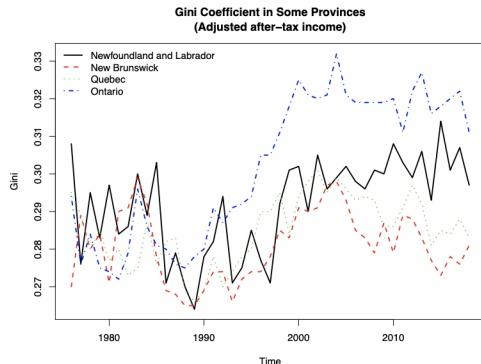
Gini coefficient in Canada



- Time series: Gini coefficient in Canada (adjusted after-tax income), 1976–2018.
- Stable through late 1980s; rises in early 1990s to early 2000s; then declines.

Gini coefficient in some provinces

- Ontario is the province where inequality increased the most over the period; by 2018, New Brunswick and Quebec had the least inequality among the four shown.

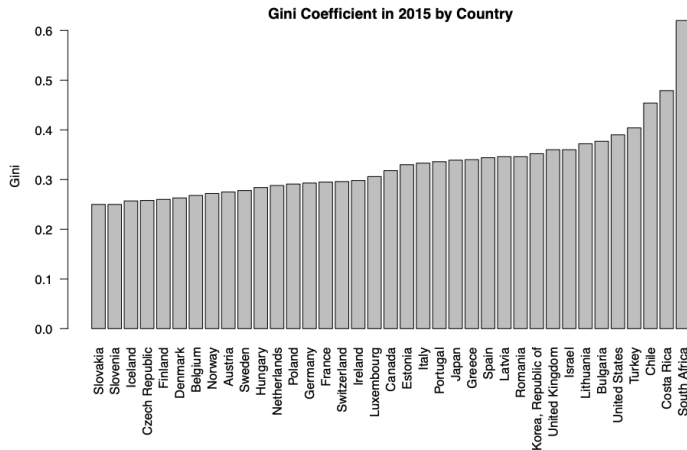


- Gini coefficient (adjusted after-tax income) for Newfoundland and Labrador, New Brunswick, Quebec, Ontario.

Comparing the Gini coefficient across countries

- To compare across countries, coefficients should come from the **same source** and be computed consistently.
- A bar chart for 2015 shows Canada in the middle among the shown countries; the United States is among the most unequal; Slovakia is among the most equal; South Africa is the most unequal in the chart.

Comparing the Gini coefficient across countries



- Bar chart: Gini coefficient in 2015 by country (selected OECD/non-OECD).

Exercises: measuring the Gini coefficient

- 1. Consider the following table of average income by quintile. Compute the Gini coefficient.

Quintile	Average Income (dollars)
First Income Quintile	342
Second Income Quintile	642
Third Income Quintile	844
Fourth Income Quintile	990
Fifth Income Quintile	1,053

Exercises: measuring the Gini coefficient (continued)

- 2. Consider the following table of average income by decile. Compute the Gini coefficient.

Decile	Average Income (dollars)
First Income Decile	114
Second Income Decile	288
Third Income Decile	605
Fourth Income Decile	621
Fifth Income Decile	1,012
Sixth Income Decile	1,023
Seventh Income Decile	1,101
Eighth Income Decile	1,434
Nineth Income Decile	1,925
Tenth Income Decile	1,934

Exercise solution (deciles): shares and cumulative shares

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- Example: for the first decile,

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- Compute cumulative shares, then apply the general formula with $n = 10$:

$$\text{Gini} = \frac{n-1}{n} - \frac{2}{n}(s_1 + s_2 + \cdots + s_9).$$

Decile shares table (rounded at the end)

Decile	Avg. Income	Income (%)	share	Cumulative shares (%)
First Income Decile	114	1.13		1.13
Second Income Decile	288	2.86		4.00
Third Income Decile	605	6.02		10.01
Fourth Income Decile	621	6.17		16.19
Fifth Income Decile	1,012	10.06		26.25
Sixth Income Decile	1,023	10.17		36.42
Seventh Income Decile	1,101	10.95		47.37
Eighth Income Decile	1,434	14.26		61.63
Ninth Income Decile	1,925	19.14		80.77
Tenth Income Decile	1,934	19.23		100.00

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 - the **market basket measure (MBM)**, and
 - the **low income measure (LIM)**.

The market basket measure (MBM)

Definition

The Market Basket Measure (MBM) is based on the cost of a specific basket of goods and services representing a modest, basic standard of living. It includes the costs of food, clothing, shelter, transportation and other items for a reference family. These costs are compared to the disposable income of families to determine whether or not they fall below the poverty line.

- A household is below the poverty line if it cannot afford the “modest and basic standard of living” determined by the basket.

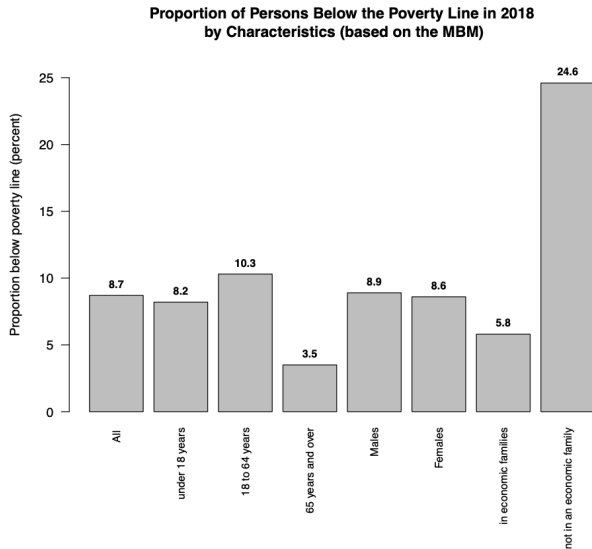
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- A household is below the poverty line if it cannot afford the “modest and basic standard of living” determined by the basket.
- The table includes the proportion of persons below the poverty line by characteristics starting in 2006.

Proportion below poverty line in 2018 (MBM)



The low income measure (LIM)

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- It is equal to half the adjusted median household income (adjusted for household size).

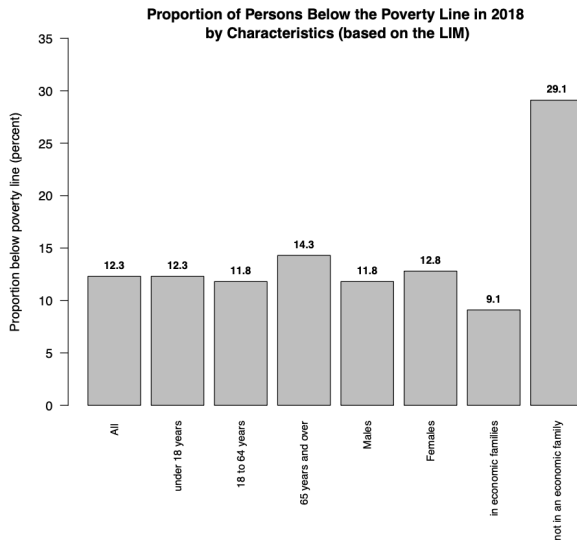
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- Median household income in 2018 was \$61,400, so households with income less than \$30,700 are below the poverty line (before adjusting for household size).
- The LIM series is based on adjusted after-tax income.

Proportion below poverty line in 2018 (LIM)



Poverty over time in Canada (LIM)

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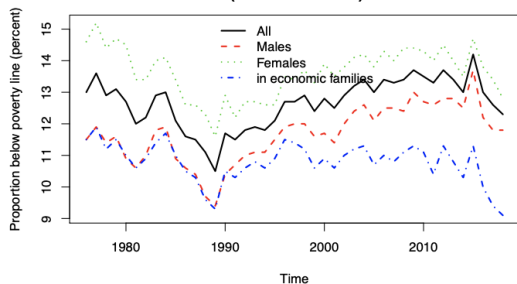
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- Overall: poverty decreased from 1976 to 1990 and increased after 1990; persons in family were quite stable.

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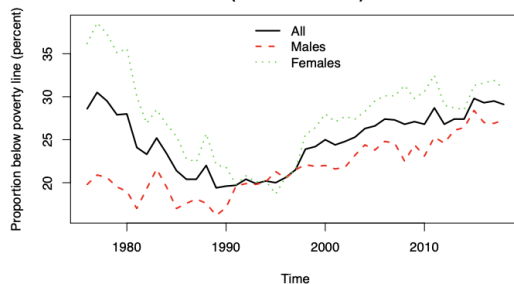
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- Overall: poverty decreased from 1976 to 1990 and increased after 1990; persons in family were quite stable.
- Females were systematically more likely to be below the poverty line than males, but the gap narrowed by 2018.

Poverty over time

**Proportion of Persons Below the Poverty Line by Characteristics
(based on the LIM)**



**Proportion of Persons Not in Family Below the Poverty Line
(based on the LIM)**



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- Everyone could be above the poverty line $\Rightarrow$ poverty low but inequality could still be high.
- Poverty focuses on the bottom of the distribution; inequality focuses on dispersion across the whole distribution.

Exercise: poverty versus per capita GDP in Canada

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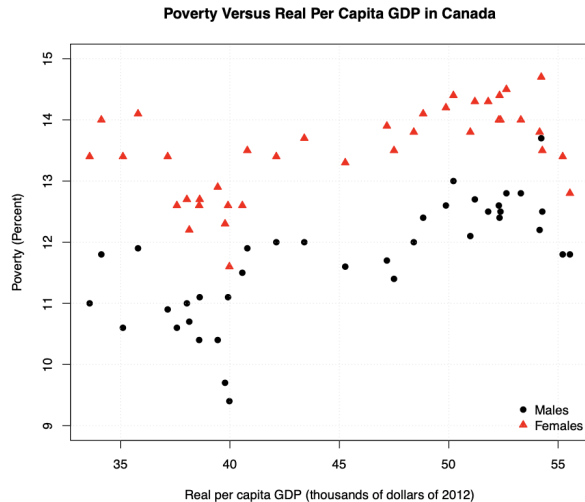
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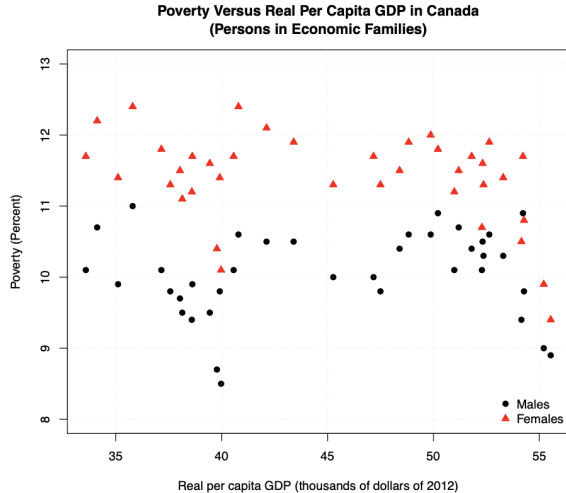
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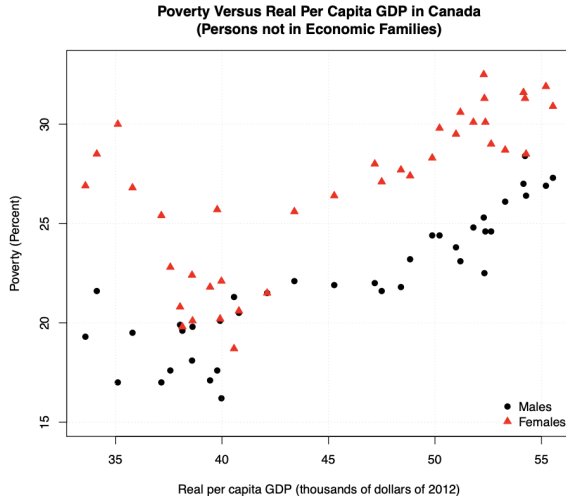
Exercise solution: scatter plots



Poverty vs GDP: persons in economic families



Poverty vs GDP: persons not in economic families



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- **Technological advances:** Often increase the value of certain skills/education (e.g., IT boom), affecting inequality.
- **Value of education:** Anything that raises the return to education amplifies how education inequality translates into income inequality.

Median annual earnings by education (2016)

	High school diploma	Apprenticeship certificate	College diploma	Bachelor's degree
Women	\$43,254	\$38,230	\$48,599	\$68,342
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- Policies that reduce education inequality can also reduce income inequality.

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- Understand the intuition behind channels and the difficulty of measuring their relative importance.

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- A key advantage of a highly mobile economy is that everyone has a more equal chance of being successful.

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- Poverty depends on the poverty line definition (MBM vs LIM) and varies by characteristics and over time.
- Inequality can interact with growth and mobility through multiple channels.